4	Water is ejected at a speed of 0.5 m s^{-1} onto a wall from the nozzle of a hose with a diameter of 0.01 m. The density of water is 1000 kg m^{-3}. If the water does not rebound, what is the force exerted by the water on the wall?							
	A	5.0 x 10^{-3} N	**B**	2.0 x 10^{-2} N	**C**	2.5 x 10^{-2} N	**D**	7.9 x 10^{-2} N
L2	**Answer: B** Force on water by wall = Momentum **change** per unit time for the **water** = (mass per unit time) x (change in velocity of water) = (density x volume per unit time) x (change in velocity of water) = (density x cross-sectional area x speed) x (final velocity – initial velocity of water) = $(1000)(\pi(0.005)^2)(0.5) \times (0 - 0.5)$ = - 0.0196 N = - 2.0 x 10^{-2} N (2 s.f.) By Newton's third law, Force on water by wall = - Force on wall by water							